



ST BENEDICT'S

SUBJECT	PHYSICS FINAL EXAM
GRADE	11
EXAMINER	Mr Hilder
MODERATOR	Mr McCoy
DATE	21 November 2022
MARKS	200
DURATION	180 minutes

MARK
TOTAL

LEARNER NAME: **MEMO**

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 30 pages and a Data Sheet of 2 pages, please detach the Data Sheet. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
4. Use the data and formulae sheet whenever necessary.
5. Show your working in all calculations.
6. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
7. Answers must be expressed in decimal format, not left as proper fractions.
8. Where appropriate, express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.
10. Blank pages are included at the end of the paper (page 29 & 30). If you run out of space for a question, use this page. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	9	10	11	12	Total
Mark													
Moderated Mark													
Question Total	20	18	17	14	15	12	20	11	17	10	28	18	200

Comment:

QUESTION 1 MULTIPLE CHOICE**20 MARKS**

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	C	D
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Here the option C has been marked as an example

1.1	A✓✓✓	B	C	D
1.2	A✓✓✓	B	C	D
1.3	A	B	C	D✓✓✓
1.4	A	B✓✓✓	C	D
1.5	A	B✓✓✓	C	D
1.6	A	B	C	D✓✓
1.7	A✓✓✓	B	C	D
1.8	A	B✓✓✓	C	D
1.9	A	B	C✓✓✓	D
1.10	A	B✓✓✓	C	D

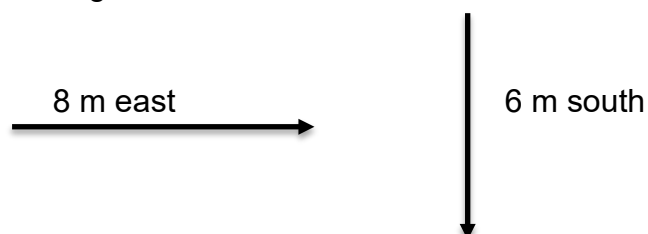
1.1 Kinetic energy can be expressed mathematically as $E_k = \frac{1}{2}mv^2$.

This equation consists of:

(2)

- A one vector quantity and two scalar quantities
- B two vector quantities and one scalar quantity
- C three vector quantities
- D three scalar quantities

1.2 The diagram below shows two vectors.



What is the magnitude of the resultant of these two vectors?

(2)

- A 10 m
- B 12 m
- C 14 km
- D 16 km

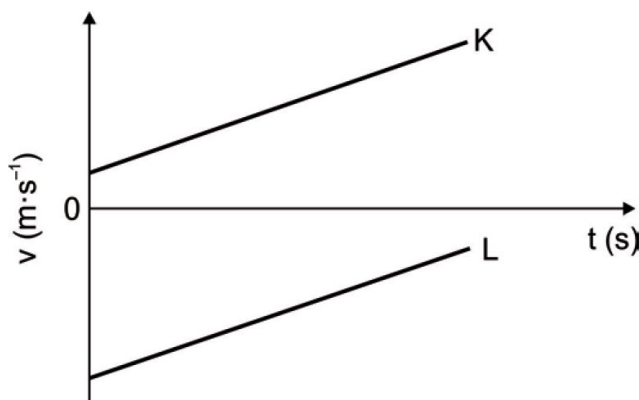
- 1.3 A car is moving with a velocity $\mathbf{v}$. The driver slams on brakes and the car stops after traveling a certain distance $\mathbf{d}$.
If the driver was traveling at triple the initial velocity, $\mathbf{3v}$ and braked with the same force, how would the car's stopping distance compare to the first distance? (2)

- A $\frac{1}{3}d$
B $3d$
C $6d$
D $9d$

- 1.4 A person stands on a bathroom scale in a stationary elevator. The reading on the scale is **490 N**. When the elevator is in motion, the reading on the scale changes to **470 N**. Which one of the following combinations best describe the direction of the motion and the direction of the acceleration of the elevator during the motion? (2)

	Direction of motion	Direction of acceleration
A	Upwards	Upwards
B	Downwards	Downwards
C	Upwards	Downwards and then upwards
D	Downwards	Upwards and then downwards

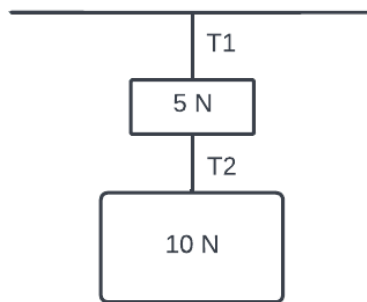
- 1.5 The velocity vs time graph shows the motion of two different objects sliding in a straight line across a horizontal surface.



For which object could friction be the cause of the change in velocity? (2)

- A K
B L
C L and K
D Neither

- 1.6 Two blocks of weight 5 N and 10 N are attached to two ropes, T1 and T2 as shown below.



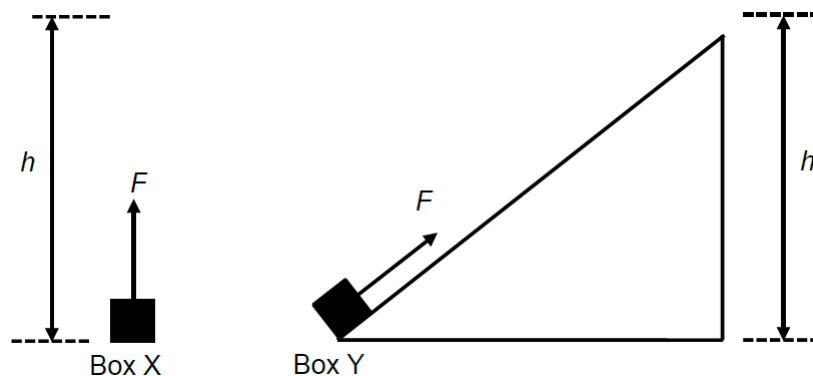
The blocks hang vertically at rest. The masses of the ropes are negligible. What is the tension in rope T1 and in rope T2?

(2)

	Tension in T1	Tension in T2
A	15 N	15 N
B	10 N	15 N
C	5 N	15 N
D	15 N	10 N

- 1.7 Two boxes X and Y have the same mass. Box X is lifted through a vertical height h by a force of magnitude F . Box Y is pulled up a slope by a force of the same magnitude to reach the same vertical height, as shown.

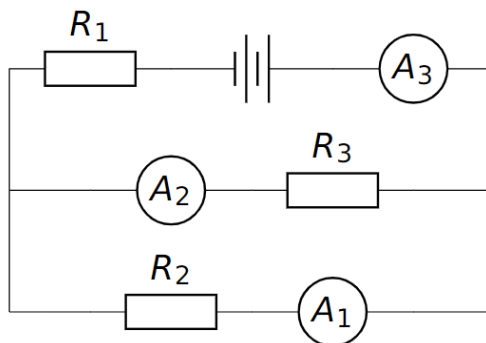
(2)



	Change in gravitational potential energy of the boxes.	Amount of work done by F.
A	Is the same for box X and Y	Is the same on X and Y
B	Is the same of box X and Y	Is more for box Y than for box X.
C	Is less for box Y than for box X	Is less for box Y than for box X
D	Is more for box Y than for box X	Is more for box Y than for box X

1.8 Which equation correctly describes the current in the circuit diagram?

(2)

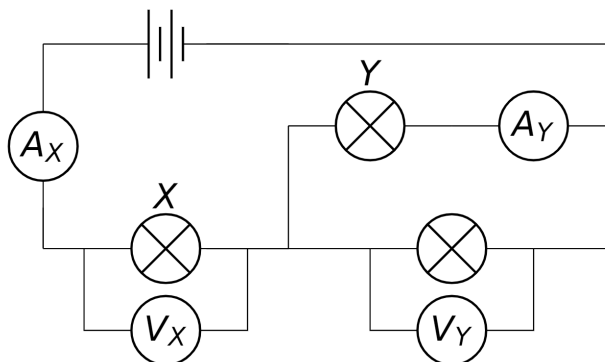


- A $A_1 = A_2 + A_3$
B $A_2 = A_3 - A_1$
 C $A_3 = A_2 - A_1$
 D $A_2 = A_1 + A_3$

1.9 In the circuit shown below the bulbs are all identical.

Which one of the following correctly gives the ratio of the power dissipated in bulb X compared to the power dissipated in bulb Y.

(2)



	P(X) : P(Y)
A	1 : 4
B	1 : 2
C	4 : 1
D	2 : 1

1.10 Which one of the following types of electromagnetic radiation has the highest energy?

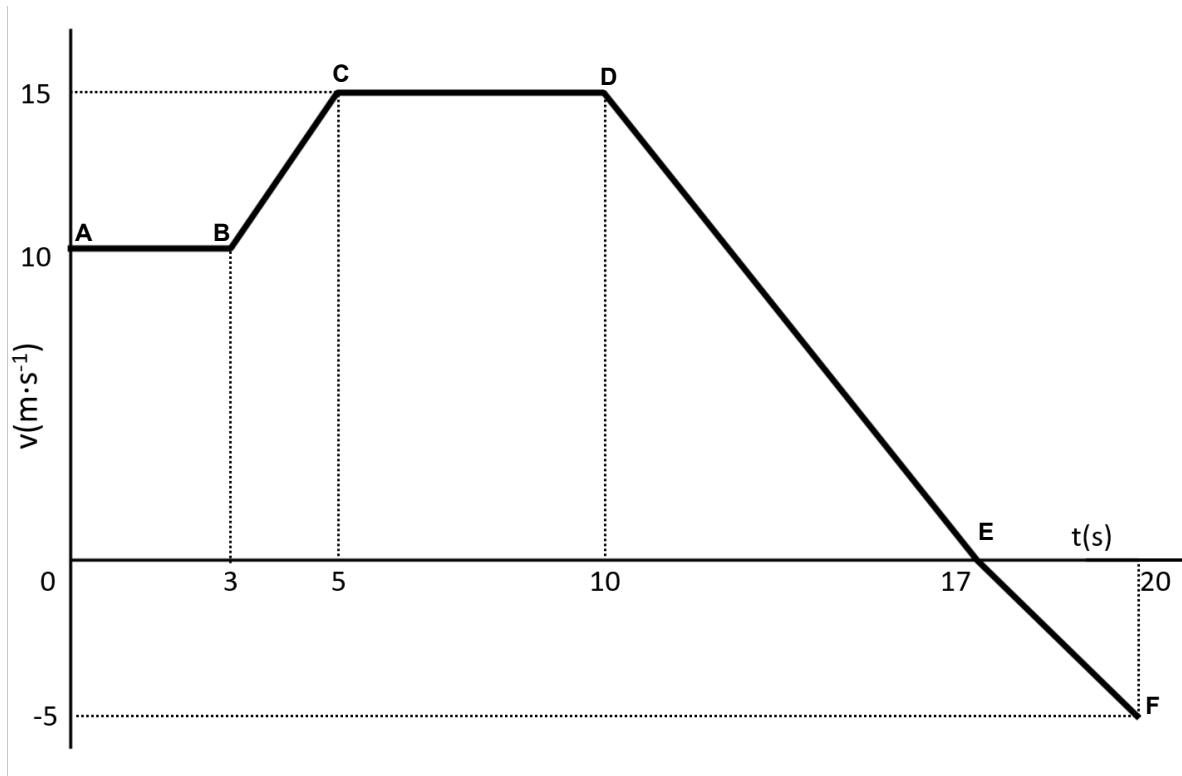
(2)

- A X-rays
B Gamma rays
 C Infra-red
 D Ultra-violet

QUESTION 2 KINEMATICS GRAPH**18 MARKS**

The motion of an object travelling south in a straight line is represented in the velocity time graph shown below.

Using the graph and without the use of the equations of motion, answer the following questions.



- 2.1 At which point(s) is the object stationary? (2)

E ✓✓

- 2.2 At which stage(s) is the object travelling with a constant velocity. (2)

AB ✓ - CD ✓

- 2.3 What is the displacement of the object between B and C? (4)

$$s = \ell b + \frac{1}{2}bh \checkmark$$

$$s = (10)(5 - 3) + \frac{1}{2}(15 - 10)(5 - 3) \checkmark$$

$$s = (10)(2) + \frac{1}{2}(5)(2)$$

$$s = 20 + 5$$

$$s = 25m \checkmark \text{ South } \checkmark$$

2.4 Calculate magnitude of acceleration between D and E.

(3)

$$m = \frac{\Delta y}{\Delta x} = \frac{v_e - v_d}{t_e - t_d} \quad \checkmark$$

$$a = \frac{0 - 15}{17 - 10} \quad \checkmark$$

$$a = \frac{-15}{7}$$

$$a = -2,14 \text{ ms}^{-2} \quad \checkmark$$

2.5 Describe the motion of the object between D and F.

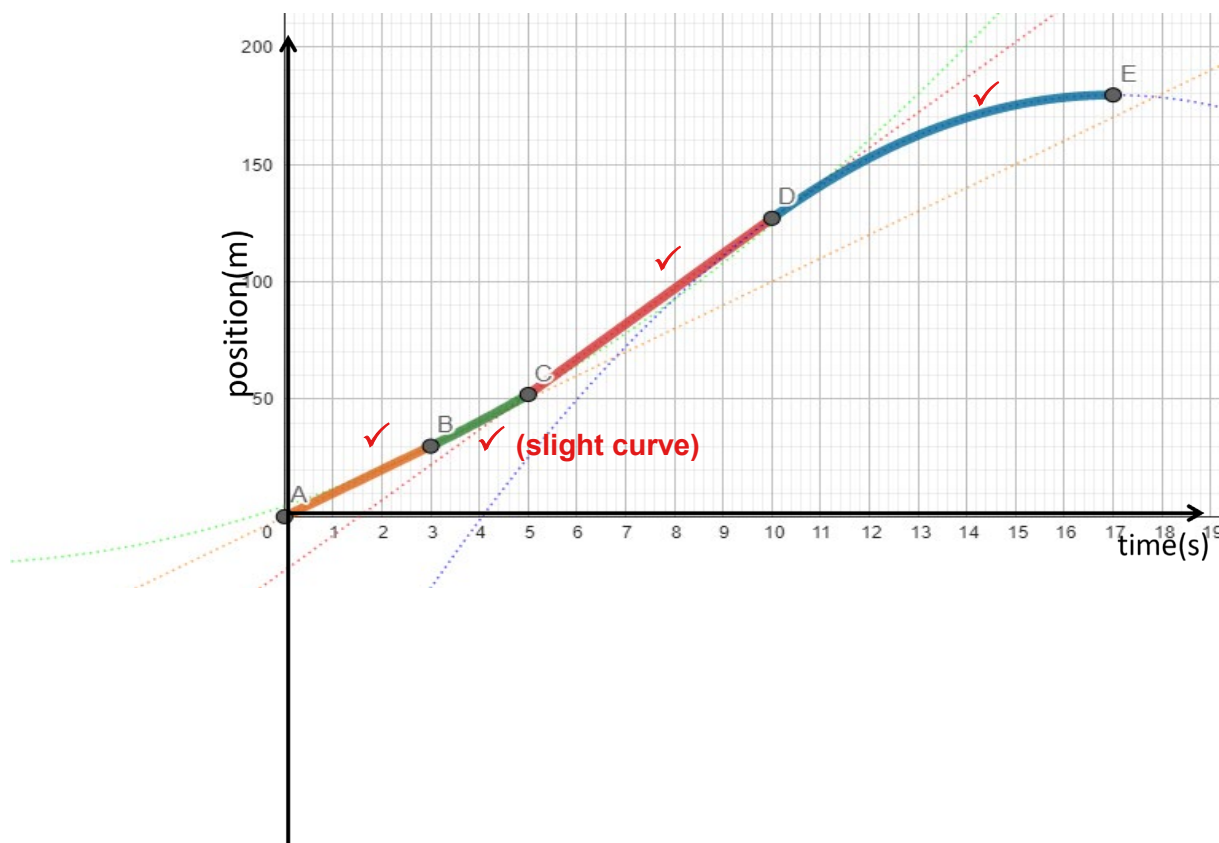
(3)

D-E : Decreasing velocity south, $\checkmark \frac{1}{2}$ constant negative acceleration to the North. $\checkmark \frac{1}{2}$

E – Object is briefly stationary $\checkmark$

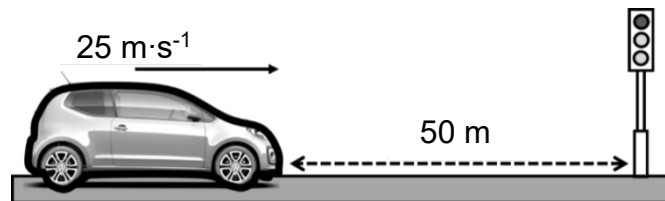
E – F : Increasing velocity northward $\checkmark \frac{1}{2}$, constant acceleration to the North. $\checkmark \frac{1}{2}$

2.6 Draw a position vs time graph of the motion between B and E on the axis provided. (4)



QUESTION 3 HORIZONTAL MOTION**17 MARKS**

A VW is travelling at a speed of $25 \text{ m} \cdot \text{s}^{-1}$ when a traffic light, 50 m ahead, changes to red. The driver takes 0,35 s to react before slamming on the brakes.



Once the brakes have been applied, the car comes to a stop 3,2 s.

3.1 Define *speed*. (2)

Rate of change of distance. ✓✓

3.2 What was the initial speed of the car in kilometres per hour? (2)

$$25 \text{ m} \cdot \text{s}^{-1} = 90 \text{ km} \cdot \text{h}^{-1} \quad \checkmark \checkmark$$

3.3 Calculate the distance the car travels during the reaction time. (2)

$$s = vt \quad \checkmark$$

$$s = (25)(0,35)$$

$$s = 8,75 \text{ m} \quad \checkmark$$

3.4 Will the car stop before the traffic light? Show ALL calculations. (4)

$$\Delta s = \frac{v + u}{2} t \quad \checkmark$$

$$s_f - s_i = \left(\frac{v + u}{2} \right) (t)$$

$$s_f - 8,75 \quad \checkmark = \left(\frac{25 - 0}{2} \right) (3,2)$$

$$s_f = (12,5)(3,2) + 8,75$$

$$s_f = 40 + 8,75$$

$$s_f = 48,75 \text{ m} \quad \checkmark$$

$\therefore$ the VW will be able to stop before the traffic light with 1,25 m to spare ✓

- 3.5 The driver's phone had been placed on the passenger seat next to them. When the brakes were applied the phone slid forward landing on the floor in the foot well. Refer to one of Newton's Laws of motion to explain why this happened. (2)

Newton's First Law. ✓

The object has inertia and the friction between the seat and the phone is not

Enough to [prevent the phone from carrying on in its motion.] ✓

- 3.6 While the WV is waiting for the traffic light to turn green, a BMW approaches from behind with a uniform velocity of $12 \text{ m}\cdot\text{s}^{-1}$. The instant the BMW is directly next to the VW, the traffic light turns green and the VW begins accelerating uniformly for 4 s reaching a velocity of $18 \text{ m}\cdot\text{s}^{-1}$. When the VW reaches $18 \text{ m}\cdot\text{s}^{-1}$ it stops accelerating and maintains this velocity. How far away from the initial traffic light will both cars be when the VW catches up with the BMW? (5)

$$\begin{aligned}
 s_{vw} &= s_{acc} + s_{v_{const}} & s_{bmw} &= s_{4s} + s_{rest} \\
 s_{vw} &= \frac{v_f + v_i}{2} t_{init}^2 + v t_x & s_{bmw} &= v t_{init} + v t_x \\
 s_{vw} &= \frac{18 + 0}{2} (4) + 18 t_x \quad \checkmark & s_{bmw} &= (12)(4) + 12 t_x \quad \checkmark
 \end{aligned}$$

$$s_{vw} = 36 + 18 t_x \quad s_{bmw} = 48 + 12 t_x$$

$$s_{vw} = s_{bmw}$$

$$36 + 18 t_x = 48 + 12 t_x$$

$$6 t_x = 12$$

$$t_x = 2 \text{ s} \quad \checkmark$$

$$t_t = t_1 + t_x$$

$$t_t = 4 + 2$$

$$t_1 = 6 \text{ s total time} \quad \checkmark$$

$$\therefore s_{bmw} = vt$$

$$s_{bmw} = (12)(6)$$

$$s_{bmw} = \boxed{72 \text{ m}} \quad \checkmark$$

QUESTION 4 PROJECTILE MOTION**14 MARKS**

A drone is being flown directly upward at a constant velocity of $3 \text{ m}\cdot\text{s}^{-1}$.

As the drone is gaining altitude the camera attached to the bottom of the drone falls off.



It takes 3,6 s to hit the ground below. Air resistance can be ignored

4.1 Define *displacement*. (2)

change in position. ✓✓

4.2 Calculate the time that it will take the camera to reach its maximum height. (3)

$$v = u + at \quad \checkmark$$

$$0 = 3 + (-9,8)t \quad \checkmark$$

$$-3 = -9,8t$$

$$t = \frac{3}{9,8}$$

$$t = 0,31 \text{ s} \quad \checkmark$$

Time to max height while moving up:

$$s = ut + \frac{1}{2}at^2$$

$$s = (3)(3,6) + \frac{1}{2}(-9,8)(3,6^2)$$

$$s = 52 \text{ m}$$

$$t = \frac{s}{v} = \frac{52,704}{3} = 17,568 \text{ s}$$

$$t_t = 17,568 + 0,31$$

$$t_t = 17,87 \text{ s}$$

4.3 At what height above the ground was the drone when the camera fell off? (3)

$$s = ut + \frac{1}{2}at^2 \quad \checkmark$$

$$s = (3)(3,6) + \frac{1}{2}(-9,8)(3,6^2) \quad \checkmark$$

$$s = 10,8 - (4,9)(12,96)$$

$$s = 10,8 - 63,504$$

$$s = -52,7 \text{ m} \quad \checkmark$$

$\therefore 52,7 \text{ m above the ground.}$

- 4.4 Calculate the maximum height reached by the camera above the ground. (4)

$$v_{max}^2 = u_{detach}^2 + 2gs_{max} \checkmark$$

$$0^2 = 3^2 + 2(-9,8)s$$

$$-9 = -19,6s$$

$$s = 0.459 \checkmark$$

$$s_{max} = s_1 + s_2$$

$$s_{max} = 0,459 + 52.7 \checkmark (c.o.e - 4.3)$$

$$s_{max} = 53,16 m \checkmark$$

- 4.5 In reality air resistance **cannot be ignored**, describe the effect air resistance would have on the distance the camera travels as it falls. (2)

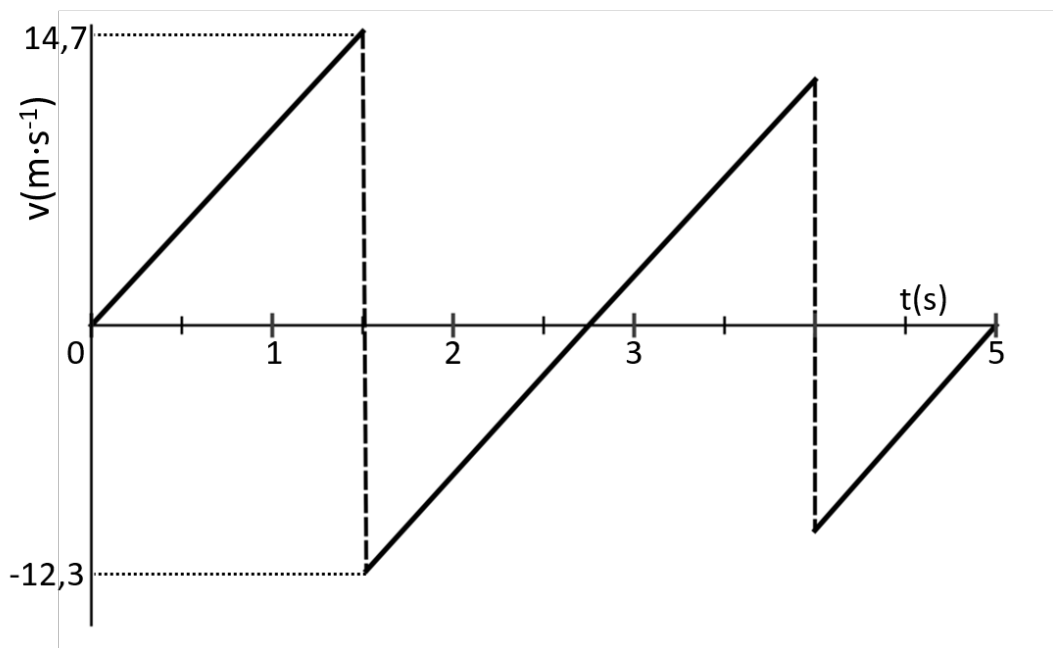
The distance would decrease $\checkmark$,

As the camera carried on moving upwards it would not be able to reach as high

as drag (due to drone and air resistance) would prevent its upward motion $\checkmark$

QUESTION 5 BOUNCE**16 MARKS**

A student drops a tennis ball from a certain height. The velocity-time graph represents the motion of the ball as it bounces vertically on a solid floor. The time of bounce is negligible and air resistance may be ignored.



5.1 Define *acceleration*. (2)

The rate of change of velocity. ✓✓

5.2 Describe the motion of the ball from 1,5 to 2,75 seconds. (2)

The ball would leave the floor with a velocity upward. ✓

It would have a negative acceleration due to gravity so would slow down

Until it reached the top of its motion. ✓

5.3 Calculate the height from which the ball was dropped. (3)

$$v^2 = u^2 + 2gs \quad \checkmark$$

$$14,7^2 = 0^2 + 2(9,8)s \quad \checkmark$$

$$216,09 = 19,6s$$

$$s = 11,03 \text{ m} \quad \checkmark$$

$$\text{area} = \frac{1}{2}bh$$

$$\text{area} = \frac{1}{2}(1,5)(14,7)$$

$$\text{area} = 11,025 \text{ m}$$

5.4 Was energy lost during the bounce? Circle **YES** ✓ or NO.

Briefly explain your answer.

(3)

The velocity decreased between hitting the ground until leaving it. ✓✓

$E_k \propto v^2$ so energy was lost.

5.5 State the magnitude of the velocity of the ball at 4 s.

(2)

12,3 ms⁻¹ ✓✓ (can accept +/- [9 m·s⁻¹] as speed it leaves ground using 70% efficiency)

5.6 Suggest a possible reason for the motion experienced by the ball at 5 s.

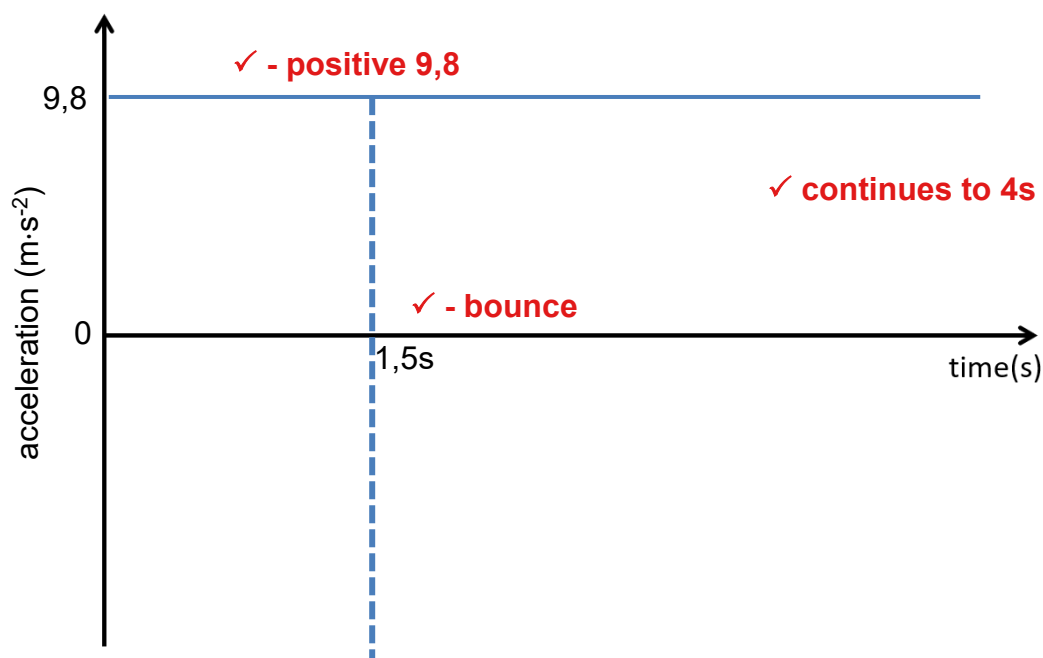
(1)

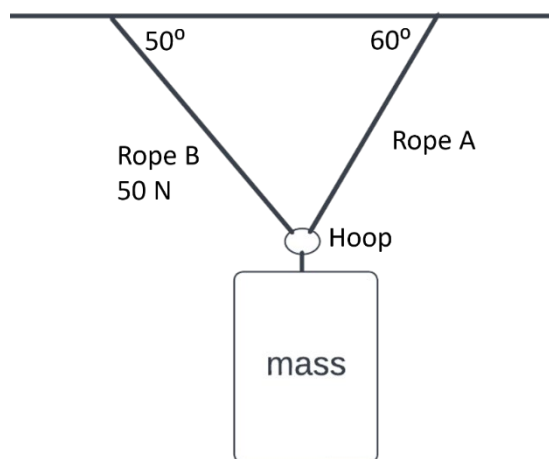
It was caught at the peak of its motion. ✓

5.7 Sketch an acceleration versus time graph of the ball from 0s to 4 s.

Indicate the acceleration of the ball when it hits the ground.

(3)



QUESTION 6 ROPES**11 MARKS**

A block is hanging at rest from a hoop, which is attached to the ceiling by two ropes. The ropes make angles to the horizontal ceiling of 50° and 60° as shown in the diagram. The tension in the rope B is 50 N.

6.1 Define a vector. (2)

A physical quantity that has both magnitude and direction. ✓✓

6.2 Determine the magnitude of the horizontal component of the tension in rope B. (2)

$$F_{xB} = F_B \cos \theta$$

$$F_{xB} = 50 \cos(50^\circ) \checkmark$$

$$F_{xB} = 32,14 \text{ N} \checkmark$$

6.3 Calculate the magnitude of the tension in the rope A. (3)

$$F_{xA} = F_{xB} \checkmark$$

$$F_{xA} = F_A \cos \theta_A$$

$$32,14 = F_A \cos(60^\circ) \checkmark$$

$$F_A = 64,23 \text{ N} \checkmark \quad (64,278 \text{ N})$$

6.4 Determine the block's mass.

(4)

$$F_{\text{net}Y} = ma$$

$$F_g + F_{yB} + F_{yA} = 0 \checkmark$$

$$-F_g = F_{yB} + F_{yA}$$

$$-(mg) = F_B \sin \theta_B + F_A \sin \theta_A$$

$$-m(-9,8) = 50 \sin(50^\circ) + 64,23 \sin(60^\circ) \checkmark$$

$$9,8m \checkmark = 38,3 + 55,67$$

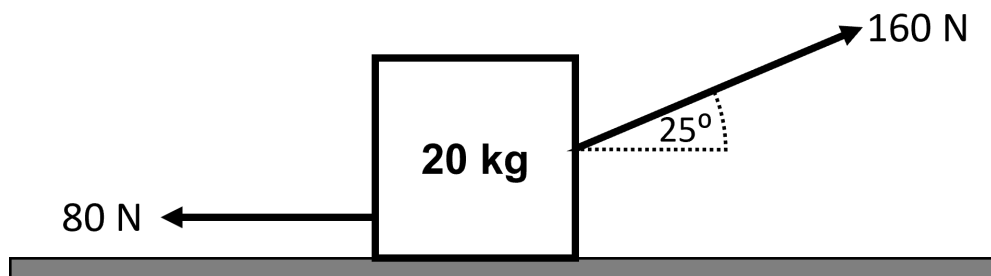
$$9,8m = 93,97$$

$$m = 9.59 \text{ kg} \checkmark$$

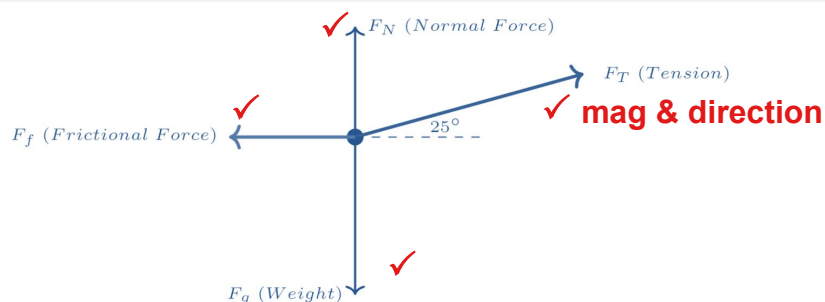
PLEASE TURN OVER

QUESTION 7 SLIDING BLOCKS**20 MARKS**

- 7.1 A 20 kg metal box is pulled across a rough horizontal surface with a force of 160 N acting at an angle of 25° to the surface. The kinetic frictional force acting on the metal box is 80 N.



- 7.1.1 Draw a free-body diagram showing all the forces acting on the metal box. (4)



- 7.1.2 State *Newton's second law of motion*. (2)

When a net force, F_{net} , is applied to an object of mass, m , it accelerates in the direction of the net force. ✓ The acceleration, a , is directly proportional to the net force and inversely proportional to the mass. ✓

- 7.1.3 Calculate the magnitude of the acceleration of the metal box. (4)

$$F_{net} = ma$$

$$F_{Tx} - F_f = ma \quad \checkmark$$

$$160 \cos(25^\circ) \checkmark - 80 = 20a \checkmark$$

$$145 - 80 = 20a$$

$$65 = 20a$$

$$a = 3,25 \text{ ms}^{-2} \quad \checkmark$$

- 7.1.4 Calculate the magnitude of the normal force that the ground is exerting on the box. (3)

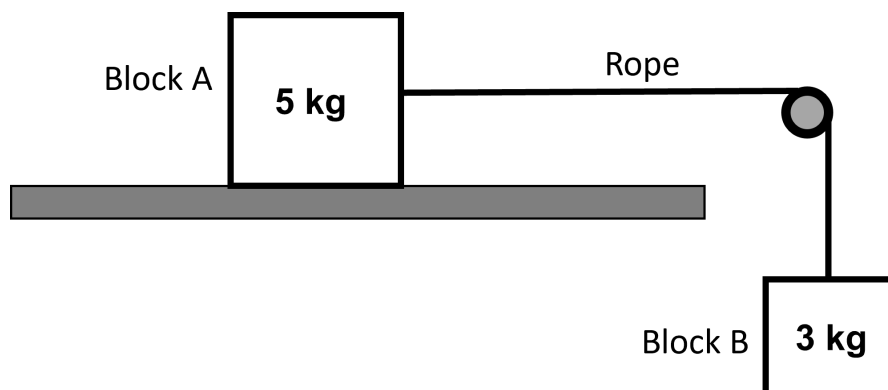
$$F_N = F_g - F_y \checkmark$$

$$F_N = (20)(9,8) - 160 \sin(25^\circ) \checkmark$$

$$F_N = 196 - 67,62$$

$$F_N = 128,38N \checkmark$$

- 7.2 Two blocks are connected by a massless rope. Block A is being pulled across a rough surface by the weight of Block B. The blocks are accelerating clockwise at $2,5 \text{ m}\cdot\text{s}^{-2}$.



- 7.2.1 What is the magnitude of friction acting on Block A? (4)

$$F_{net} = ma$$

$$F_g + F_f = ma \checkmark$$

$$(3)(9,8) \checkmark + F_f = 8(2,5) \checkmark$$

$$29,4 + F_f = 20$$

$$F_f = -9,4 N \checkmark$$

7.2.2 Determine the tension in the Rope.

(3)

$$F_{net} = ma$$

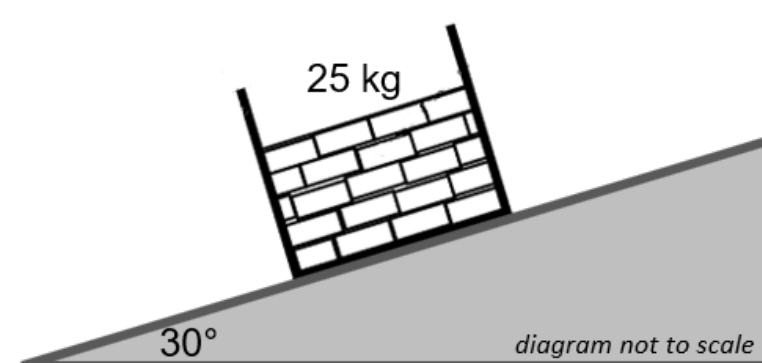
$$(3)(9,8) - F_t = (3)(2,5) \checkmark$$

$$29,4 - 7,5 = F_t \checkmark$$

$$F_t = 21,9N \checkmark$$

QUESTION 8 SLOPE**11 MARKS**

8.1 A crate of bricks with a mass of 25 kg is on the verge of slipping when placed on a wooden plane inclined at 30° from the horizontal.

8.1.1 State *Newton's first law*.

(2)

An object continues in a state of rest or uniform (moving with constant)

velocity unless it is acted upon by a net or resultant force. $\checkmark \checkmark$

8.1.2 What is the magnitude of the net force (F_{net}) acting on the container of bricks while it is at rest on the slope?

(1)

Zero (0N) $\checkmark$

8.1.3 Calculate the normal force acting on the box.

(2)

$$F_N = F_{g\parallel}$$

$$F_N = mg \cos \theta \quad \checkmark$$

$$F_N = (25)(9,8) \cos 30$$

$$F_N = 212,18N \quad \checkmark$$

8.1.4 Use you answer from QUESTION 8.1.3 to determine the co-efficient of static friction between the crate and the surface of the slope.

(3)

$$F_k = \mu_k F_N$$

$$(F_k = -F_{g\parallel} \quad \& \quad F_N = -F_{g\perp})$$

$$mg \sin \theta = \mu_k mg \cos \theta \quad \checkmark$$

$$(25)(9,8) \sin 30 \quad \checkmark = 212,18 \mu_k$$

$$\frac{122,5}{212,18} = \mu_k$$

$$\mu_k = 0.58 \quad \checkmark$$

8.1.5 What effect would lowering the incline of the slope have on the co-efficient of static friction? INCREASE , DECREASE or REMAINS THE SAME $\checkmark$. Circle the correct option.

(1)

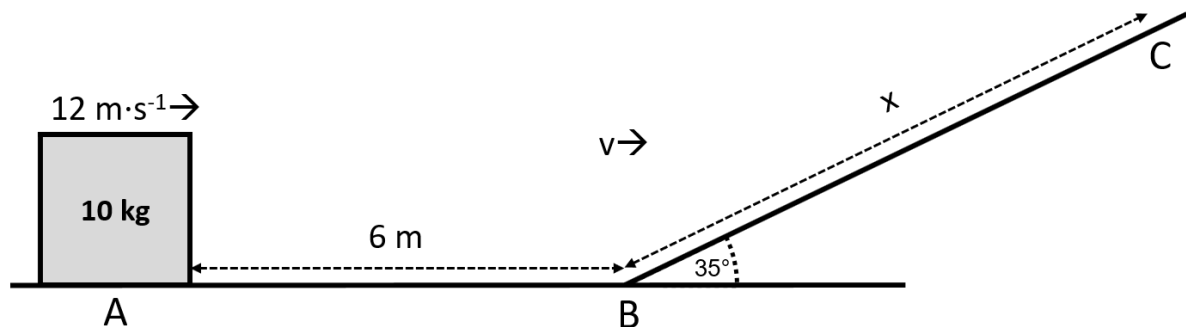
8.1.6 Explain your answer to QUESTION 8.1.5

(2)

The coefficient of friction is based on the interaction between the two surfaces $\checkmark$ The angle of the slope has no effect $\checkmark$ The ratio of $\frac{F_f}{F_N}$ remains the same as $\mu \propto F_f$ & $\mu \propto \frac{1}{F_N}$.

QUESTION 9 WORK-ENERGY**17 MARKS**

A block of mass 10 kg is sliding along surface. The horizontal surface from A to B is rough and the surface from B to C is a smooth incline at 35° from the horizontal.



The block is travelling at a speed of $12 \text{ m}\cdot\text{s}^{-1}$ as it passes A.

- 9.1 Define the term *kinetic energy*. (2)

the energy an object has as a result of the object's motion. ✓✓

- 9.2 Calculate the kinetic energy of the block as it passes A. (3)

$$E_K = \frac{1}{2}mv^2 \quad \checkmark$$

$$E_K = \frac{1}{2}(10)(12^2) \quad \checkmark$$

$$E_k = (5)(144)$$

$$E_k = 720 \text{ J} \quad \checkmark$$

The frictional force acting on the block as it slides from A to B is 54,9 N.

- 9.3 State the *work-energy theorem*. (2)

the work done by a net force on an object is equal to the change in the kinetic energy of the object – the work–energy theorem. ✓✓

9.4 Calculate the speed of the block (v) as it reaches B. (4)

$$W_{net} = \Delta E_k$$

$$F_f \Delta x \cos \theta_x = E_{k_f} - E_{k_i} \checkmark$$

$$(54,9)(6)\cos(-180) \checkmark = \frac{1}{2}mv^2 - 720 \checkmark (c.o.e - 9.2)$$

$$-329,4 + 720 = \frac{1}{2}(10)v^2$$

$$390,6 = 5v^2$$

$$v = \sqrt{78,12}$$

$$v = 8,84 \text{ ms}^{-1} \checkmark$$

The block slides up the incline from B and comes to rest at C. The frictional force acting on the block as it slides from B to C is negligible.

9.5 Define *mechanical energy* (2)

sum of gravitational potential and kinetic energy at a point. $\checkmark \checkmark$

9.6 Calculate the distance, x , that the block slides up the slope before coming to rest at C. (4)

$$E_{Mc} = E_{Mb}$$

$$E_{p_c} + E_{k_c} = E_{k_b} + E_{p_b} \checkmark$$

$$mg \sin \theta_s \Delta x \cos \theta_x \checkmark + 0 = \frac{1}{2}mv^2 + 0$$

$$(10)(-9,8) \cdot \sin(35) \cdot \Delta x \cdot \cos(-180) + 0 = \frac{1}{2}(10)(8,84^2) \checkmark + 0 (c.o.e - 9.4)$$

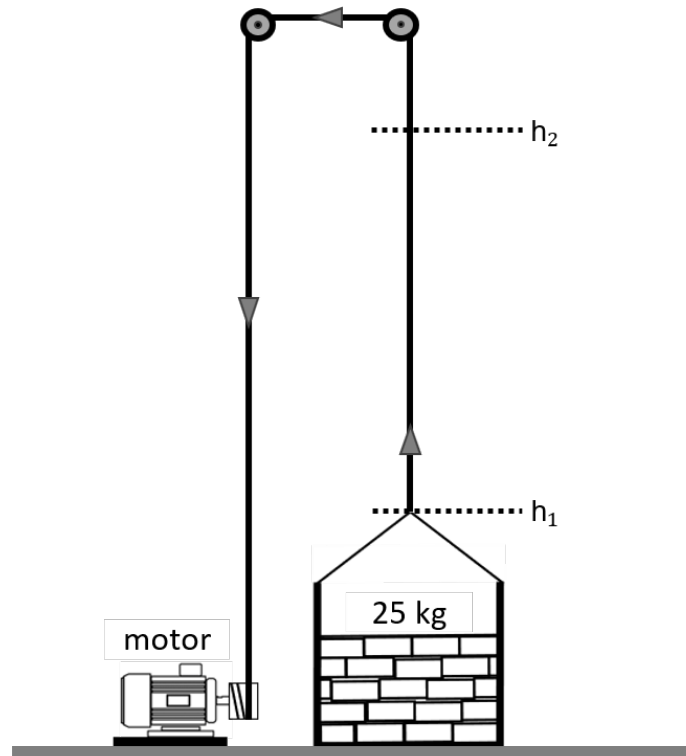
$$56,21 \Delta x = (5)(78,126)$$

$$\Delta x = \frac{390,728}{56,21}$$

$$\Delta x = 6,95 \text{ m} \checkmark$$

QUESTION 10 POWER**10 MARKS**

A 1600 Watt electric motor is used to lift a 25 kg box of bricks from h_1 (0 m) to h_2 (18 m) at a constant velocity.



10.1 Calculate the power required to lift the bricks at a constant velocity of $6 \text{ m}\cdot\text{s}^{-1}$. (3)

$$P = Fv$$

$$P = F_g v \checkmark$$

$$P = (mg)v$$

$$P = (25)(9,8)(6) \checkmark$$

$$\boxed{P = 1470 \text{ W} \checkmark}$$

10.2 Determine the work done by the motor to lift the box 18 m at the stated velocity. (3)

$$E_p = mgh \checkmark$$

$$E_p = (25)(9,8)(18) \checkmark$$

$$\boxed{E_p = 4410 \text{ J} \checkmark}$$

10.3 How efficient is the motor while lifting the box of bricks at $6 \text{ m}\cdot\text{s}^{-1}$?

(4)

$$efficiency = \frac{P_{out}}{P_{in}} \times 100 \checkmark$$

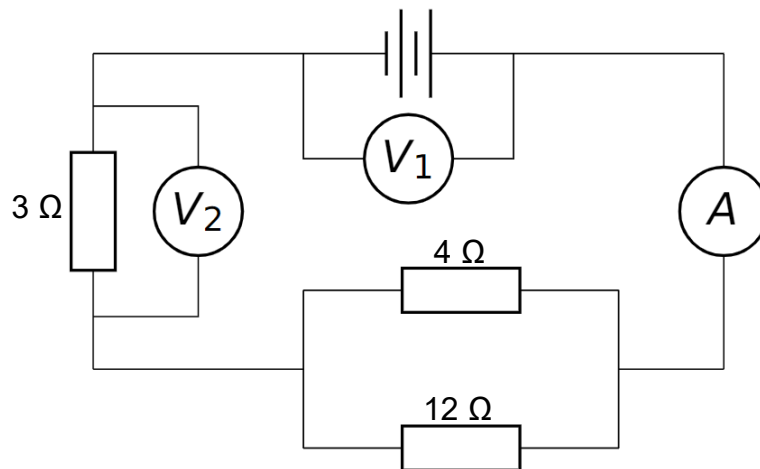
$$efficiency = \frac{1470 \checkmark (c.o.e - 10.1)}{1600 \checkmark} \times 100$$

$$efficiency = 91,88\% \checkmark$$

PLEASE TURN OVER

QUESTION 11 CIRCUITS**28 MARKS**

- 11.1 A circuit that consists of a battery with an emf of 18 V, three resistors, an ammeter and two voltmeters are set up as shown in the diagram below.



- 11.1.1 Define *resistance*. (2)

Opposition to flow of current. ✓✓

- 11.1.2 Calculate the effective resistance of the circuit. (4)

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_p} = \frac{1}{12} + \frac{1}{4} \quad \checkmark$$

$$R_p = 3\Omega \quad \checkmark$$

$$R_T = R_s + R_t \quad \checkmark$$

$$R_T = 3 + 3$$

$$R_T = 6\Omega \quad \checkmark$$

The ammeter measures a current of 3 A with the circuit connected as shown.

- 11.1.3 State *Ohm's Law*. (2)

Current through a conductor is directly proportional to ✓

the potential difference across the conductor at constant temperature. ✓

11.1.4 Determine the current through the 4 Ω resistor.

(4)

$$V_2:V_1$$

$$9:9 \checkmark$$

$$I = \frac{V}{R} \checkmark$$

$$I = \frac{9}{4} \checkmark$$

$$I = 2,25A \checkmark$$

11.1.5 Determine the power dissipated by the 3 Ω resistor.

(3)

$$P = \frac{V^2}{R} \checkmark$$

$$P = \frac{9^2}{3} \checkmark$$

$$P = 27 W \checkmark$$

Sometimes resistors burn out while in operation11.1.6 Describe what would happen to the reading on the voltmeter, V_2 , if the 12 Ω resistor burnt out. Use a suitable equation in your answer.

(3)

R_t would increase as it is now a series circuit. $\checkmark$
This would cause less current to flow. $\checkmark$
As $V \propto I$ this means the voltage would decrease . $\checkmark$

11.1.7 Considering the 12 Ω resistor has burnt out, how much energy would the battery supply to the circuit in 2 minutes?

(4)

$$W = \frac{V^2}{R} \times t \checkmark$$

$$W = \frac{18^2}{7} \checkmark \times 120 \checkmark$$

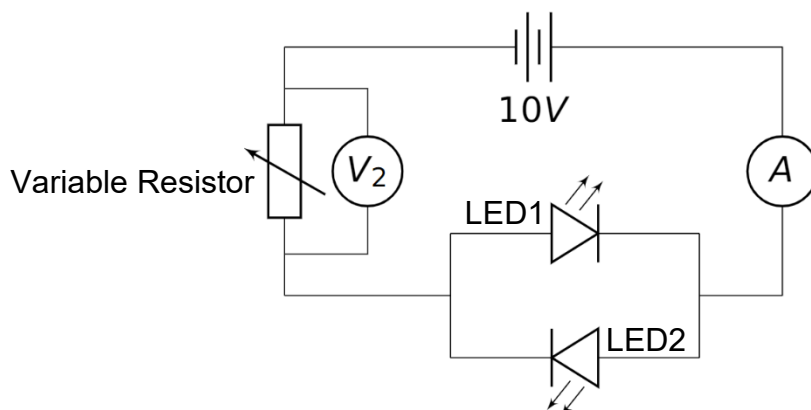
$$W = 5554,29 J \checkmark$$

$$W = VIt \checkmark$$

$$W = (18)(2,57 \checkmark)(120 \checkmark)$$

$$W = 5554,29 J \checkmark$$

- 11.2 A new circuit is built using two LEDs and a variable resistor as shown in the diagram. Both LEDs have a knee voltage of 2,5 V at a current of 20 mA.



- 11.2.1 Which LED will not light up? **LED1** ✓ or LED2.

Circle the correct option.

(1)

- 11.2.2 Explain your answer to QUESTION 11.2.1.

(1)

It is connected in reverse bias ✓

- 11.2.3 What should the resistance of the variable resistor be set to, in order to provide enough voltage for the LED to operate.

(4)

$$V_1 = 10 - 2,5 = 8,5 \text{ V} \checkmark$$

$$I = 20 \text{ mA} = 0,02 \text{ A} \checkmark$$

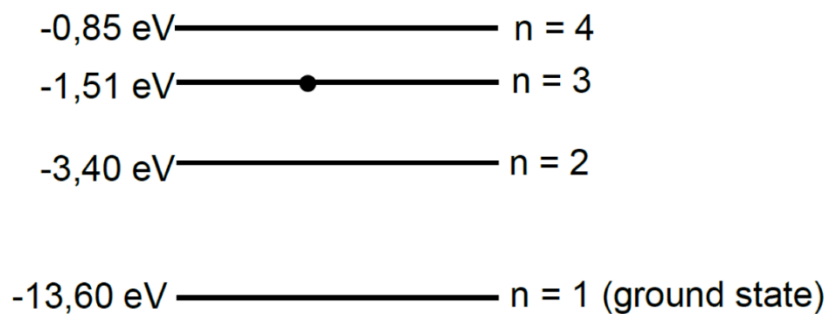
$$R = \frac{V}{I} \checkmark$$

$$R = \frac{8,5}{0,02}$$

$$R = 375 \Omega \checkmark$$

QUESTION 12 WAVES AND LIGHT**18 MARKS**

12.1 The energy levels for a hydrogen atom are represented on the diagram below. The energy levels for a hydrogen atom are represented on the diagram below.



12.1.1 An excited electron is in the 3rd energy level as indicated.

How many unique frequencies will be in the emission spectrum as the excited electron returns to the ground state? (1)

3✓

12.1.2 Calculate the frequency of the photon emitted when the electron transition is from the $n=2$ level to the ground state. (4)

$$\Delta E = 13,6 - 3,4 \checkmark$$

$$\Delta E = (10,2)1,6 \times 10^{-19} = 1,632 \times 10^{-18} \checkmark$$

$$E = hf \checkmark$$

$$1,632 = 6,6 \times 10^{-34} f$$

$$f = 2,47 \times 10^{15} \text{ Hz} \checkmark$$

Hydrogen gas in a discharge tube emits light of wavelength 655 nm.

12.1.1 Calculate the energy (in eV) that corresponds to this wavelength of light. (4)

$$E = \frac{hc}{\lambda} \checkmark$$

$$E = \frac{(6,6 \times 10^{-34})(3 \times 10^8)}{655 \times 10^{-9}}$$

$$E = 3,02 \times 10^{-19} \text{ J} \checkmark$$

$$\text{eV} = \frac{3,02 \times 10^{-19}}{1,6 \times 10^{-19}} \checkmark$$

$$\text{eV} = 1,89 \text{ eV} \checkmark$$

PLEASE TURN OVER

12.1.2 Which transition that will result in a wavelength of 655 nm? (2)

$$3 - 2 \checkmark \checkmark$$

12.1.3 Why are the energy levels stated as negative values? (2)

The negative indicates the amount of energy that needs to be absorbed

to move into free space. $\checkmark \checkmark$

12.2 A FM radio station broadcasts using waves with a wavelength of 3,17 m.

12.2.1 What is the frequency of their transmission? (3)

$$c = f\lambda \checkmark$$

$$3 \times 10^8 = f (3,17) \checkmark$$

$$f = 94,63 \times 10^6 \text{ Hz} \checkmark$$

702 Talk Radio broadcasts using AM radio waves which have a frequency in the kHz.

12.2.2 Refer to the properties of FM and AM radio waves to explain why AM is not suitable for music-based radio but can be used for talk radio. (2)

The lower frequency means less information can be transmitted. $\checkmark$

The higher frequency is required to send information with music which contains

A large amount of data. $\checkmark$

END OF ASSESMENT

ADDITIONAL SPACE

[illegible]

